Random Walks and Markov Chains

Friday seminar

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Outline

1. Review
2. Convergence of Random Walks on Undirected Graphs
    - ε-mixing time
3. Electrical Networks and Random Walks
    - Harmonic functions
    - The analogy between electrical networks and random walks
4. Random Walks on Undirected Graphs with Unit Edge Weights
    - Hitting time, commute time, cover time
5. Random Walks in Euclidean Space
    - Escape probability
6. The Web as a Markov Chain
    - Pagerank

Notation

- $p(t)$: A row vector of probability distribution after t steps of a random walk.
$$p(t)P = p(t+1),$$
where $P = [p_{xy}]$ is a transition matrix of probability of going from vertex x to y.
- Big O: We write $f(n) = O(g(n))$ if and only if there exists a positive constant C such that
$$f(n) \leq Cg(n) \quad \text{as } n \to \infty.$$
- Big Ω: We write $f(n) = \Omega(g(n))$ if and only if there exists a positive constant c such that
$$f(n) \geq cg(n) \quad \text{as } n \to \infty.$$
- Big Θ: We write $f(n) = \Theta(g(n))$ if and only if there exist positive constant C_1 and C_2 such that
$$C_1 g(n) \leq f(n) \leq C_2 g(n) \quad \text{as } n \to \infty.$$

Stationary Distribution

Long-term average probability distribution $a(t)$ is defined by

$$a(t) = \frac{1}{t}(p(0) + p(1) + \cdots + p(t-1)).$$

Fundamental Theorem of Markov Chains

For a connected Markov chain there is a unique probability vector π satisfying $\pi P = \pi$. Moreover, for any starting distribution, $\lim_{t\to\infty} a(t)$ exists and equals π.

Lemma 4.3

For a random walk on a strongly connected graph with probabilities on the edges, if the vector π satisfies $\pi_x p_{xy} = \pi_y p_{yx}$ for all x and y and $\sum_x \pi_x = 1$, then π is the stationary distribution of the walk.

Convergence of Random Walks on Undirected Graphs

- Let w_{xy} be the weight of the edge between the vertices x and y and $w_x := \sum_y w_{xy}$.
- The transition probability is given by $p_{xy} = \frac{w_{xy}}{w_x}$.
- The stationary distribution π is given by

$$\pi_x = \frac{w_x}{w_{tot}}, \quad w_{tot} := \sum_x w_x.$$

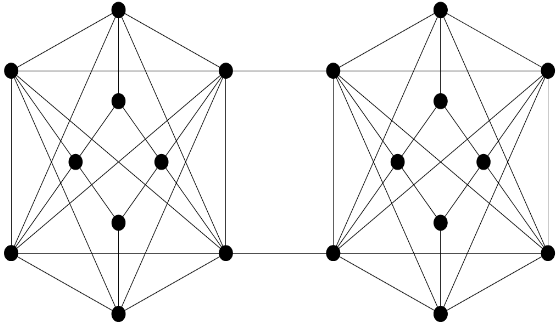
$$\because \quad w_x p_{xy} = w_x \frac{w_{xy}}{w_x} = w_{xy} = w_{yx} = w_y \frac{w_{yx}}{w_y} = w_y p_{yx},$$

which yields $\pi_x p_{xy} = \pi_y p_{yx}$ and Lemma 4.3 implies that π is the stationary distribution.

What is the time by which the average probability distribution is guaranteed to be close to the stationary distribution?
$\rightsquigarrow$ *ε-mixing time*

Definition 4.1

Let $\varepsilon > 0$. The ε-mixing time of a Markov chain is the minimum integer t such that for any starting distribution p, the ℓ_1-distance between the t-step running average probability distribution and the stationary distribution is at most ε.



A graph with large ε-mixing time

Definition 4.2

For a subset S of vertices, let $\pi(S)$ denote $\sum_{x \in S} \pi_x$. The normalized conductance $\Phi(S)$ of S is

$$\Phi(S) = \frac{\sum_{(x,y) \in (S,\bar{S})} \pi_x p_{xy}}{\min\{\pi(S), \pi(\bar{S})\}}.$$

(*Heuristic interpretation*) WLOG, $\pi(S) \leq \pi(\bar{S})$.

- $\pi(S) = \sum_{x \in S} \underbrace{\frac{\pi_x}{\pi(S)}}_{a} \underbrace{\sum_{y \in \bar{S}} p_{xy}}_{b}$.

  a: probability of being in x if we were in the stationary distribution,
  b: probability of stepping from x to $\bar{S}$ in a single step.
  $\therefore \pi(S) =$ probability of moving from S to $\bar{S}$ in a single step if we are in the stationary distribution restricted to S.

- Using $\pi_x = w_x / w_{tot}$, we have

$$\Phi(S) = \frac{1}{w_{tot}\pi(S)} \sum_{(x,y) \in (S,\bar{S})} w_{xy}.$$

$\approx \#$ of edges leaving S to $\bar{S}$

ε-mixing time

Definition 4.3

The normalized conductance of the Markov chain, denoted by Φ, is defined by

$$\Phi := \min_{S \subset V, S \neq \phi} \Phi(S).$$

Theorem 4.5

The ε-mixing time of a random walk on an undirected graph is

$$O\left(\frac{\log(1/\pi_{min})}{\Phi^2 \varepsilon^3}\right),$$

where π_{min} is the minimum stationary probability of any state.

High conductance $\Longrightarrow$ rapid mixing.

1D case

- $p_{xy} = \frac{1}{2}$ and $\pi_x = \frac{1}{n}$ for all x.
- The set S minimizing Φ, i.e., with the smallest $\Phi(S) = \frac{\sum_{(x,y)\in(S,\bar{S})} \pi_x p_{xy}}{\min\{\pi(S),\pi(\bar{S})\}}$, i.e., the smallest number of edges leaving it is the first $n/2$ vertices. In this case, $\Phi(S) = \frac{1}{n}$.
- By Theorem 4.5, the ε-mixing time is $O(n^2 \log n/\varepsilon^3)$.
- Also for $d \geq 2$, the ε-mixing time is $O(n^2 \log n/\varepsilon^3)$.

Electrical Networks and Random Walks

Random Walks on Undirected Graphs with Unit Edge Weights

In this section, we study the relationship between *electrical networks* and *random walks* on undirected graphs.

- Assign a resistance $r_{xy} > 0$ to each edge (x, y) and define conductance $c_{xy} := \frac{1}{r_{xy}}$.
- Consider the underlying graph with $p_{xy} = \frac{c_{xy}}{c_x}$ where $c_x := \sum_y c_{xy}$.
- Note that p_{xy} is not necessarily equal to p_{yx}.
- Fundamental Theorem of Markov Chains yields there exists a unique stationary probability distribution π. Indeed,

$$\pi_x = \frac{c_x}{c_0}, \quad c_0 := \sum_x c_x.$$

To see this, for all x and y,

$$\pi_x p_{xy} = \frac{c_x}{c_0}\frac{c_{xy}}{c_x} = \frac{c_y}{c_0}\frac{c_{yx}}{c_y} = \pi_y p_{yx},$$

and Lemma 4.3 implies that π is the unique stationary probability.

Harmonic functions

Definition

Let $B \subset V$ be given(we call B boundary). A harmonic function g on the graph is a function whose value on B is fixed to some boundary condition, and the value at the interior vertices($\bar{B}$) satisfies the following property:

$$g_x = \sum_y g_y p_{xy}.$$

- A harmonic function on a connected graph takes on its maximum and minimum on the boundary.
$\Big(\because$ let S be the set of vertices where the maximum value is attained and assume that S contains no boundary vertices. There exists an edge (x, y) such that $x \in S, y \in \bar{S}$. Since g_x is a maximum and $g_x > g_y$, we have

$$g_x = \sum_z g_z p_{xz} < g_x \sum_z p_{xz} = g_x,$$

which is a contradiction. The same argument can be applied for the minimum. $\Big)$
- A harmonic function is unique if the boundary condition is given.

The analogy between electrical networks and random walks

- Attach a voltage source between the vertices a and b such that $v_a = 1, v_b = 0$, which induces a current flow through the edges.

- We can give probabilistic interpretation of voltage, current and resistance.

Probabilistic interpretation of voltages

1. The voltage v_x is a harmonic function.

By Ohm's Law,

$$\underbrace{i_{xy}}_{\text{current } x \to y} = \frac{v_x - v_y}{r_{xy}} = (v_x - v_y)c_{xy}.$$

By Kirchhoff's law,

$$\sum_y i_{xy} = 0.$$

$$\implies \sum_y (v_x - v_y)c_{xy} = 0, \quad \text{i.e.,} \quad v_x \sum_y c_{xy} = \sum_y v_y c_{xy}.$$

Using $\sum_y c_{xy} = c_x$ and $p_{xy} = \frac{c_{xy}}{c_x}$, we have

$$v_x = \sum_y v_y p_{xy}.$$

Probabilistic interpretation of voltages

2. $v_x = \mathbb{P}(\text{a walk starting at } x \text{ reaching } a \text{ before } b)$.

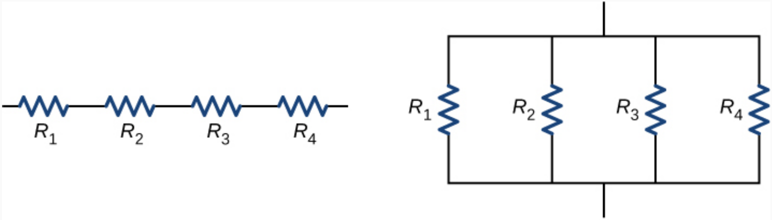
Denote by q_x the right-hand side. Then,

- $q_a = 1 = v_a$ and $q_b = 0 = v_b$.
- $q_x = \sum_y p_{xy} q_y$, i.e., q_x is harmonic.

$$\implies \quad v_x = q_x.$$

Effective resistance and escape probability

- i_a: the current flowing into the network at a and out at b.
- The effective resistance $r_{eff} := \frac{v_a - v_b}{i_a}$, the effective conductance $c_{eff} := \frac{1}{r_{eff}}$.
- r_{eff} can be regarded as single resistor.



(a) Resistors connected in series (b) Resistors connected in parallel

$$(a) \quad r_{eff} = \sum R_i, \qquad\qquad (b) \quad \frac{1}{r_{eff}} = \sum \frac{1}{R_i}.$$

Effective resistance and escape probability

- The escape probability
  $p_{escape} := \mathbb{P}(\text{a walk starting at } a \text{ reaches } b \text{ before returning to } a)$.

Then, we have

$$p_{escape} = \frac{c_{eff}}{c_a}.$$

To see this, Ohm's law yields $i_a = \sum_y (v_a - v_y) c_{ay}$. Since $v_a = 1$,

$$\begin{aligned} i_a &= \sum_y c_{ay} - c_a \sum_y v_y \frac{c_{ay}}{c_a} \\ &= c_a \Big(1 - \sum_y p_{ay} v_y\Big). \end{aligned}$$

Observe that $\sum_y p_{ay} v_y = \mathbb{P}(\text{a walk starting at } a \text{ returning to } a \text{ before reaching } b)$. So, $1 - \sum_y p_{ay} v_y = \mathbb{P}(\text{a walk starting at } a \text{ reaching } b \text{ before returning to } a) = p_{escape}$ and $i_a = c_a p_{escape}$. Using $c_{eff} = \frac{i_a}{v_a} = i_a$, we conclude the result.

Random Walks on Undirected Graphs with Unit Edge Weights

Random Walks on Undirected Graphs with Unit Edge Weights

In this section, we consider random walks on undirected graphs with uniform edge weights, i.e., for each x, $p_{xy} = p_{xz} \ \forall y, z \in \mathcal{N}(x)$.

We are interested in

1. Hitting time: What is the expected time it takes from x to y?

2. Commute time: What is the expected time it takes to return to the starting point?

3. Cover time: What is the expected time to reach every vertex?

Hitting time

The hitting time is the expected time of a random walk from x to y.

Lemma 4.7

The expected time for a random walk starting at one end of a path of n vertices to reach the other end is $\Theta(n^2)$.

Proof: Let $h_{i,j}, i < j$ be the hitting time of reaching j starting from i. Then, $h_{1,2} = 1$ and

$$h_{i,i+1} = \frac{1}{2} + \frac{1}{2}(1 + h_{i-1,i+1}) = 1 + \frac{1}{2}(h_{i-1,i} + h_{i,i+1}), \quad 2 \leq i \leq n-1,$$

so we have

$$h_{i,i+1} = 2 + h_{i-1,i}.$$

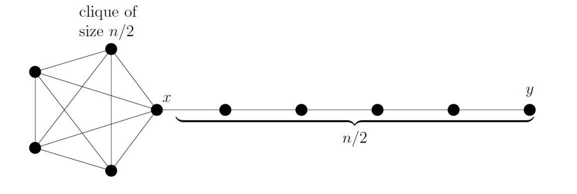
It readily gives $h_{i,i+1} = 2i - 1$ and therefore

$$h_{1,n} = \sum_{i=1}^{n-1} h_{i,i+1} = (n-1)^2.$$

Hitting time

- If G is a complete graph K_n, then $h_{i,j} = \Theta(n)$.
  $\because$ $h_{i,j}$ is the expectation of geometric distribution, i.e., the distribution of number of trials until the first success with the probability of success being $\frac{1}{n-1}$, so $h_{i,j} = n - 1$.
- Adding edges to a graph can either increase or decrease the hitting time.
- Consider a *Lollipop graph* by making the first $n/2$ vertices a complete graph $K_{\frac{n}{2}}$.

Hitting time



- $h_{x,y} = \Theta(n^3)$.
  Let w and i denote the vertex in the complete graph and the path, respectively. Denote $k := n/2$.

$$h_{x,y} = \frac{k-1}{k} h_{w,y} + \frac{1}{k}(h_{1,y} + 1),$$
$$h_{w,y} = \frac{k-2}{k-1} h_{w,y} + \frac{1}{k-1}(h_{x,y} + 1),$$
$$h_{i,y} = \frac{1}{2} h_{i-1,y} + \frac{1}{2}(h_{i+1,y} + 1).$$

- However, $h_{y,x} = \Theta(n^2)$.
- So, adding edges to a path increased the hitting time from $\Theta(n^2)$ to $\Theta(n^3)$. Adding edges again to form K_n decrease the hitting time to $\Theta(n)$.

Commute time

- The commute time, $\text{commute}(x, y)$, is the expected time of a random walk starting at x reaching y, and then returning to x.
- $\text{commute}(x, y) = h_{xy} + h_{yx}$, so it is symmetric.
- Highly related to effective resistance.

Commute time

Theorem 4.9

Given a connected, undirected graph, consider the electrical network where each edge of the graph is replaced by a one ohm resistor. Given vertices x and y, we have

$$\mathrm{commute}(x, y) = 2mr_{xy},$$

where r_{xy} is the effective resistance from x to y and m is the number of edges in the graph.

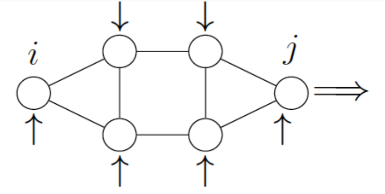
Corollary 4.10

If the x and y are connected by an edge, then $\mathrm{commute}(x, y) \leq 2m$.

Corollary 4.11

For vertices x and y in an n vertex graph, $\mathrm{commute}(x, y) \leq n^3$.

Commute time



(Sketch of proof)

- At each vertex i, insert a current $d_i A$, where $d_i = deg(i)$. Total current inserted is $2m$.
- Extract from specific vertex j, $2mA$.
- Denote by v_{ij} the voltage difference between i and j.
- For each k, adjacent to i, Ohm's law yields the current through the resistor between i and k is $v_{ij} - v_{kj}$.

$$d_i = \sum_{k \in \mathcal{N}(i)} (v_{ij} - v_{kj}) = d_i v_{ij} - \sum_{k \in \mathcal{N}(i)} v_{kj},$$

Commute time

$$v_{ij} = 1 + \sum_{k \in \mathcal{N}(i)} \frac{1}{d_i} v_{kj} = \sum_{k \in \mathcal{N}(i)} \frac{1}{d_i}(1 + v_{kj}).$$

- By definition of hitting time h_{ij},

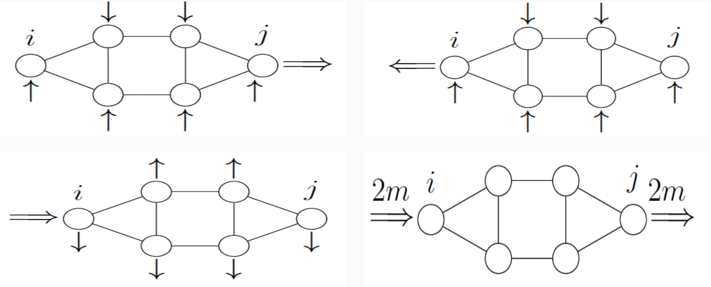
$$h_{ij} = \sum_{k \in \mathcal{N}(i)} \frac{1}{d_i}(1 + h_{kj}).$$

- Then,

$$v_{ij} - h_{ij} = \sum_{k \in \mathcal{N}(i)} \frac{1}{d_i}(v_{kj} - h_{kj}) \Longrightarrow v_{ij} - h_{ij} \text{ is harmonic.}$$

- Designate $\{j\}$ the boundary. Since the function is 0 at j, it is 0 everywhere, i.e., $v_{ij} = h_{ij}$. So, voltage difference = hitting time.

Commute time



- Repeat the previous argument with i and j interchanged.(2nd fig) Then, the new voltage satisfies $v_{ji} = h_{ji}$.
- Reverse all currents in the previous step.(3rd fig) Then, the new voltage satisfies $-v_{ji} = h_{ji}$. So, $v_{ij} = -v_{ji} = h_{ji}$.
- Superposition of 1st and 3rd circuit yields 4th circuit. We have

$$2mr_{ij} = v_{ij} = h_{ij} + h_{ji}.$$

Commute time

Corollary 4.10

If the x and y are connected by an edge, then $\text{commute}(x, y) \leq 2m$.

Since the effective resistance is bounded by the effective resistance of the shortest path(non-trivial),

$$h_{xy} + h_{yx} = 2mr_{xy} \leq 2m.$$

Corollary 4.11

For vertices x and y in an n vertex graph, $\text{commute}(x, y) \leq n^3$.

For the same reason,

$$h_{xy} + h_{yx} = 2mr_{xy} \leq 2\binom{n}{2}(n-1) \leq n^3.$$

Cover time

- The cover time, $\text{cover}(x, G)$, is the expected time of a random walk starting at vertex x in the graph G to reach each vertex at least once. The cover time of a graph G, denoted by $\text{cover}(G)$, is

$$\text{cover}(G) = \max_x \text{cover}(x, G).$$

- (Example)

$$\text{cover}(G) = \begin{cases} \Theta(n^2), & G = \text{path} \\ \Theta(n^3), & G = \text{Lollipop} \\ \Theta(n \log n), & G = K_n. \end{cases}$$

Cover time

If $G = K_n$, then it is called the *coupon collector problem*, which is

> *Q: If one wants to collect all the n types of coupons, how many coupons should he buy?*
> *A:* $\Theta(n \log n)$.

Because...

Let $T(r) = \mathbb{E}[\#$ of steps until r distinct vertices have been visited for the first time]. Note that $T(1) = 0$, $T(2) - T(1) = 1$. Suppose that $r < n$ distinct vertices have been visited for the first time. Then, the time to visit any vertex out of those unvisited $n - r$ vertices follows the geometric distribution with the parameter $\frac{n-r}{n-1}$. The expectation of it is $\frac{n-1}{n-r} = T(r+1) - T(r)$. Thus,

$$\begin{aligned}
\mathrm{cover}(G) &= T(n) - T(n-1) + T(n-1) - T(n-2) + \cdots + (T(2) - T(1)) + \\
&= \sum_{r=1}^{n} \frac{n-1}{n-r} = \Theta(n \log n).
\end{aligned}$$

Cover time

The following are some of the bounds for cover time.

- For any graph G with n vertices,

$$\mathrm{cover}(G) \geq (1 + o(1))n \log n,$$

  where the lower bound is achieved by K_n.

- For a connected graph G with n vertices and m edges,

$$\mathrm{cover}(G) \leq 4m(n-1).$$

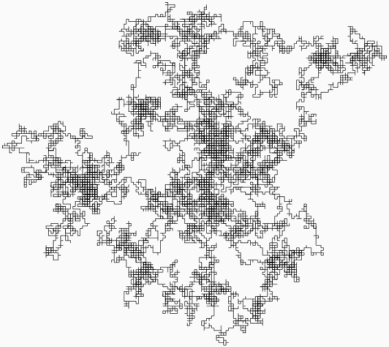
- For an undirected graph G with n vertices and m edges,

$$mr_{\mathit{eff}}(G) \leq \mathrm{cover}(G) \leq 6emr_{\mathit{eff}}(G) \log n + n,$$

  where $r_{\mathit{eff}}(G) := \max_{xy}(r_{xy})$.

Random Walks in Euclidean Space

Random Walks in Euclidean Space



Random walk in $\mathbb{R}^2$

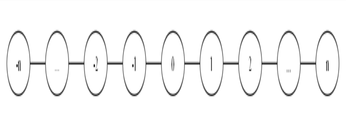
- A random walk in Euclidean space is just a special case of random walk on graphs.
- What is the probability of a random walk in $\mathbb{R}^d$ starting from the origin returning back?

Random Walks in Euclidean Space

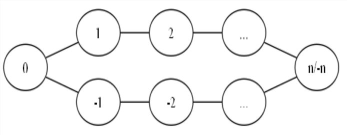
- Associate each edge with a one ohm resistor.
- Recall that the probability of a walk starting at 0 reaching the boundary before returning to 0 is the escape probability

$$p_{escape} = \frac{c_{eff}}{c_a} = \frac{1}{2dr_{eff}}.$$

- For d=1, after *shorting* the vertices n and $-n$, the electric network is two series connections of n one ohm resistors connected in parallel.



(a) 1D graph



(b) After shorting n and $-n$

Random Walks in Euclidean Space

Shorting: vertex identification. Cutting: edge deletion.

(Some facts)

- Shorting decreases r_{eff}.
- Cutting increases r_{eff}.
- If two vertices x and y have the same electrical potential and are not connected by a resistor, then shorting x and y does not change the voltage/current/resistance in the circuit.

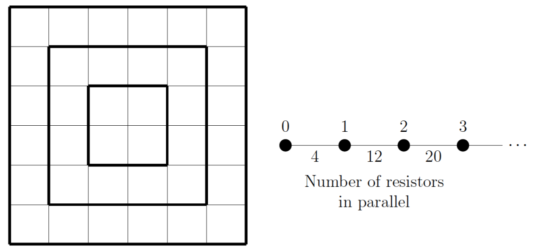
Random Walks in Euclidean Space

In this case,

$$r_{eff} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{n}{2} \to \infty, \quad \text{so } p_{escape} \to 0 \quad \text{as } n \to \infty.$$

- Note that $\lim_{n\to\infty} p_{escape}$ is the probability that the random walk starting from 0 will never return.
- Thus, the walk in the unbounded 1D lattice will return to the starting point with probability 1.

Random Walks in Euclidean Space



(a) (b)

- Similarly, for $d = 2$, we short the vertices of each square. Then, we have

$$r_{eff} \geq \frac{1}{4} + \frac{1}{12} + \frac{1}{20} + \cdots = \frac{1}{4}\sum_{i=0}^{\infty} \frac{1}{2i+1} = \Theta(\log n).$$

- Therefore, $p_{escape} \to 0$ as $n \to \infty$.

Random Walks in Euclidean Space

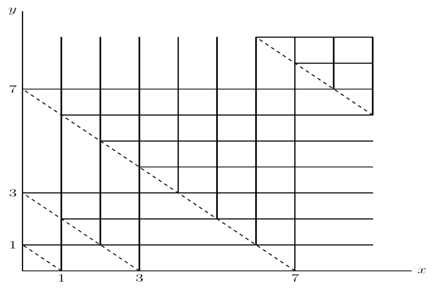
- However, for $d \geq 3$ it becomes different.
- Consider $d = 3$, for example. We now short the cubes.
- Since there are $6(2n+1)^2$ edges between nth cube and $(n+1)$th cube, we have

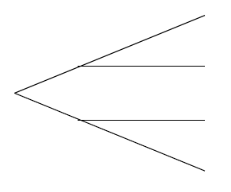
$$r_{eff} \geq \frac{1}{6}\left(1 + \frac{1}{9} + \frac{1}{25} + \cdots\right) = \frac{1}{6}\sum_{i=0}^{\infty} \frac{1}{(2i+1)^2} \geq \frac{1}{5}.$$

- Therefore, $p_{escape} = \frac{1}{2dr_{eff}} \leq \frac{5}{6}$.

What about the upper bound of r_{eff}?

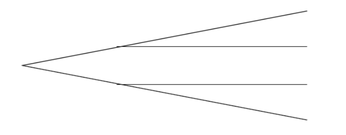
Random Walks in Euclidean Space

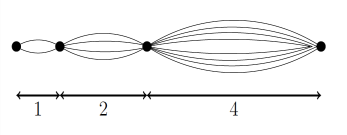




- For simplicity, consider $d = 2$.
- Dotted lines represent $x + y = 2^n - 1$.
- Perform cutting.
- Each path splits when it meets a dotted line.
- Due to the symmetry, we can split each vertex of intersection into two.

Random Walks in Euclidean Space





- Perform shorting. Then the effective resistance is

$$r_{eff} \leq \frac{1}{2} + \frac{1}{4}2 + \frac{1}{8}4 + \cdots = \infty.$$

- It doesn't work! However, for $d = 3$ it converges:

$$r_{eff} \leq \frac{1}{3} + \frac{1}{9}2 + \frac{1}{27}4 + \cdots = \frac{1}{3}(1 + \frac{2}{3} + \frac{4}{9} + \cdots) = 1.$$

- So, $p_{escape} = \frac{1}{2dr_{eff}} \geq \frac{1}{6}$. $\therefore$ $p_{escape} \in [\frac{1}{6}, \frac{5}{6}]$.

Random Walks in Euclidean Space

In summary,

$$\mathbb{P}(\text{random walk starting from the origin will return}) = \begin{cases} 1 & d = 1, 2, \\ (0, 1) & d \geq 3. \end{cases}$$

The Web as a Markov Chain

Pagerank

- An application of random walks on directed graph for establishing the importance of pages on the World Wide Web.
- Devised by Sergey Brin and Larry Page in 1998. They wanted search listings to always list the“good stuff” on the top.

Founders of Google

Pagerank

In their paper *"The Anatomy of a Large-Scale Hypertextual Web Search Engine"*, they wrote

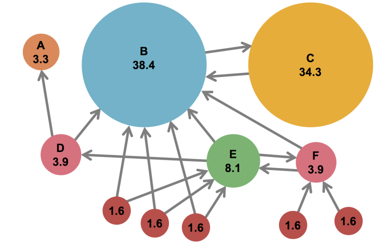
> *Our main goal is to improve the quality of web search engines. In 1994, some people believed that a complete search index would make it possible to find anything easily. (우리의 주요 목표는 검색 엔진의 품질을 향상시키는 것입니다. 1994년 당시, 사람들은 검색 인덱스를 완성하고 나면 무엇이든 쉽게 찾을 수 있을 것이라고 생각했습니다.)*
>
> *However, the Web of 1997 is quite different. Anyone who has used a search engine recently, can readily testify that the completeness of the index is not the only factor in the quality of search results. "Junk results" often wash out any results that a user is interested in. (하지만, 1997년의 웹은 사뭇 다릅니다. 최근에 검색 엔진을 사용해 본 사람이라면 누구나 인덱스를 완성하는 것만으로는 좋은 품질의 검색 결과를 얻을 수 없다는 것을 압니다. '쓰레기 정보'가 종종 사용자들이 진정 관심있어하는 정보를 가려버립니다.)*

Pagerank!

Pagerank

- Search engines output an ordered list of webpages in response to users' search query.
- To do this, they perform the following procedures.
  1. Find the set of all webpages containing the query term.
  2. Rank the webpages and display them.

  Search Engine creates a list of all the webpages(billions...) according to pagerank beforehand. As soon as the users search terms, they present the list of webpages containing the terms.



Pagerank

Pagerank

- WWW $\longleftrightarrow$ graph, URL $\longleftrightarrow$ vertex, hyperlink $\longleftrightarrow$ directed edge.
- Rank pages according to their stationary probability.
- The graph may not be connected. To solve this problem, they set restart probability $r = 0.15$ at each step, i.e., the probability that a user get bored at a webpage and jump to other webpage which is not directed by the previous webpage.

Pagerank

- The restart probability prevents π_i goes to infinity. To see this, consider a vertex i with a single edge in from vertex j and a single edge out:

$$\cdots j \longrightarrow i \longrightarrow \cdots$$

- Since π satisfies $\pi P = \pi$, $\pi_i = \pi_j p_{ji}$.
- If we add a self-loop at i, we have $\pi_i = \pi_j p_{ji} + \frac{1}{2}\pi_i$.
- Adding k self-loops at i yields $\pi_i = \pi_j p_{ji} + \frac{k}{k+1}\pi_i$. So,

$$\pi_i = (k+1)\pi_j p_{ji} \to \infty \quad \text{as} \quad k \to \infty.$$

- Taking the restart probability $r = 0.15$ into account, we have $\pi_i = 0.85\pi_j p_{ji} + 0.85\frac{k}{k+1}\pi_i$, so

$$\pi_i = \frac{0.85k + 0.85}{0.15k + 1}\pi_j p_{ji}.$$

Summary

1. Convergence of Random Walks on Undirected Graphs
    - ε-mixing time
2. Electrical Networks and Random Walks
    - Harmonic functions
    - The analogy between electrical networks and random walks
3. Random Walks on Undirected Graphs with Unit Edge Weights
    - Hitting time, commute time, cover time
4. Random Walks in Euclidean Space
    - Escape probability
5. The Web as a Markov Chain
    - Pagerank

THANK YOU FOR LISTENING!